

Where Does the Qur'an Sit, Stylistically, Next to Pre-Islamic Poetry?

A computational writing-style comparison, and what it does and doesn't show

A recent paper clustered 260 pre-Islamic Arabic poets by writing style, using an Arabic-language sentence-embedding model and a density-based clustering method, to see whether distinct stylistic “schools” showed up among poets from that era. We reproduced that paper’s method exactly, then extended it by adding the Qur’an into the same comparison, in a number of different ways, to see where it would land relative to poets writing in the same broad era and language.

This note walks through what we did and what came out of it, in the order we actually learned it, including the results that complicated our own earlier conclusions. Nothing here is about authorship, and nothing here can settle a question about authorship. It measures writing style as captured by one specific computational method, and that’s all it measures.

Method, briefly

Every poet, and every version of the Qur’an text, gets turned into a single numeric vector using an Arabic sentence-embedding model. For a poet, that means embedding each verse, averaging the verses in a poem to get a poem vector, then averaging a poet’s poems to get one vector for that poet. The same hierarchical averaging was applied to the Qur’an at several different levels: each of the 114 chapters on its own, the 60 traditional “parts” (hizb) the text is divided into, 30 pairs of consecutive parts, and the entire text averaged into one single vector. Once everything is a vector, we either cluster a whole group of them together using UMAP and HDBSCAN, the same two-step method the original paper used, or, when there are only two vectors being compared, we just report their direct cosine similarity, since clustering doesn’t mean much with only two points.

Does the Qur'an separate cleanly from poetry at all?

The first thing we tried was the simplest version of the question. We took the entire Qur’an, averaged into one vector, and dropped it into the same space as the 260 poets, 261 entities in total. It did not form its own island. It landed inside an existing cluster alongside 19 poets, with a similarity to its five closest poets ranging from 0.971 to 0.976. For comparison, poets in this corpus average about 0.799 similarity to each other, so this is a notably tight fit.

We also tried the reverse framing: instead of one combined Qur’an vector, we treated each of the 114 chapters as its own separate entity and clustered all of them alongside the 260 poets, 374 entities in total. About a quarter of the

chapters, 26 percent, ended up sharing a cluster with poets rather than clustering only with other chapters.

Neither of these results settles anything on its own, but together they meant the Qur'an wasn't simply and completely separating itself from this poetry corpus in the most basic version of the test. That's what led to everything that follows.

Does it matter which poets we're comparing against?

A natural worry: is this really about the Qur'an's style, or is it an artifact of how much text a given poet happens to have surviving? A poet with only one or two short poems produces a thin, less distinctive style signature, and thin signatures tend to sit closer to the middle of the space, close to almost anything, the Qur'an included, just because they don't have enough data to look like anything specific.

We tested this two ways. First, we kept only the 213 poets whose entire surviving output is roughly the size of a single chapter or smaller, and reran the chapter-by-chapter comparison. The overlap went up, not down: 39 percent of chapters ended up sharing a cluster with a poet, and the overall clustering got noticeably cleaner in the process. Then we ran the opposite filter, keeping only the 35 poets whose surviving output is larger than the biggest single chapter, and reran the same test. This time the overlap dropped to zero. Every one of the 114 chapters clustered only with other chapters; none shared a cluster with any poet.

Poet pool	Chapters sharing a cluster with a poet	Clustering quality (silhouette)
All 260 poets	26%	0.469
213 minor poets only	39%	0.607
35 well-documented poets only	0%	0.679

That's a clean, symmetric pattern: less poet data, more overlap with the Qur'an; more poet data, no overlap at all. It's exactly what you'd expect if thin, sparse style signatures are what's pulling minor poets close to almost anything, rather than something specific about how the Qur'an is written.

It wasn't quite that clean everywhere, though. We reran the whole-Qur'an-as-one-vector comparison, not the by-chapter one, restricted to just the 35 well-documented poets. The Qur'an still landed inside a cluster, this time with 9 poets instead of 19, at a similarity of 0.967, barely below the 0.976 from the full 260-poet version. So the "more poet data, no overlap" pattern held cleanly when the Qur'an was split into 114 separate chapters, but only partly when it was treated as one combined vector. That gap turned out to matter, and we come back to it below.

Is this about style, or just about being alone in the room?

Before drawing conclusions from individual chapters joining poet clusters, we needed to check something more basic about how this clustering method behaves. A cluster needs several similar points sitting close together to form at all. A single new point, dropped into a space with nowhere else similar to it, can't create its own cluster; it either gets pulled into whatever's nearest, or it's marked as not fitting anywhere. We needed to know which of those two outcomes was more like a coin flip and which was closer to automatic.

We tested this by dropping one random chapter at a time into the 35-poet space, ten separate times with ten different random chapters. Nine times out of ten, it landed inside a poet cluster. We repeated the test with a random bundle of ten chapters merged into one entity, again ten times: this time it happened every single time.

That result alone was suggestive but not conclusive, so we ran a control. Instead of a real chapter, we generated ten batches of genuinely meaningless text: real Arabic words drawn from the corpus vocabulary, shuffled into random order, carrying no actual content or style at all. We dropped each one into the same 35-poet space the same way the real chapters had been dropped in. It got absorbed into a poet cluster in all ten trials, at least as often as the real chapters had been.

That changes how the earlier numbers should be read. A single unfamiliar point, real Qur'anic text or pure nonsense, gets pulled toward whatever's nearby almost automatically; the clustering method doesn't have anywhere else to put it. This doesn't prove there's no real similarity between the Qur'an and this particular group of poets. It does mean that a lone chapter joining a poet cluster isn't, by itself, evidence of anything, since meaningless text does the same thing at the same rate.

Combining all the poets into one voice

A different way to ask the same question: instead of comparing many individual poets, or one lone Qur'anic unit against many poets, what happens if all the poets are collapsed into a single combined vector, the same way the Qur'an gets collapsed into one whole-text vector, and the two are compared directly?

We tried this three ways: averaging all 260 poets into one vector, averaging just the 35 well-documented poets, and averaging the 19 poets who shared the Qur'an's cluster back in the very first test.

Poet group averaged together	Similarity to whole Qur'an
All 260 poets	0.954
35 well-documented poets	0.962
19 poets from the original cluster	0.972

All three came back very high, and the number barely moves whether the average is built from a huge, maximally diverse set of 260 poets or a small, specific set of 19. That consistency points somewhere specific: heavily averaging a lot of separate texts into a single vector seems to itself produce a more generic, centrally-located result, more or less regardless of exactly which texts go into the average. That's a property of averaging as a mathematical operation, not necessarily a property of who or what is being averaged. It gives a second, independent explanation for why the whole-Qur'an-as-one-vector version of the test kept showing a stronger connection to poetry than the by-chapter version did: collapsing 114 chapters into a single vector may smooth away exactly the differences that let individual chapters separate cleanly.

We also checked the 9 poets from the well-documented cluster individually, one at a time, instead of averaged together. All 9 came back in a narrow band, 0.965 to 0.967, so no single poet was doing outsized work; the closeness was spread evenly across the whole small group rather than driven by one outlier.

Turning the test around

If a lone Qur'anic unit gets pulled into a poet cluster mostly because it's alone, the same thing should happen in reverse: a single combined poet-vector, dropped among many separate Qur'anic units, should also get pulled in rather than standing apart, regardless of any real underlying similarity.

We checked this nine separate ways: the all-260 average, the 35-poet average, and the 19-poet average, each tested against the 114 chapters, the 60 traditional parts, and the 30 part-pairs, one at a time. In every one of these nine tests, the poet-vector landed inside some cluster. It never once came back as a standalone outlier. That matches the synthetic-text result exactly: a lone point, whatever it actually contains, tends to get pulled in rather than standing apart, and that holds in both directions.

What the Qur'an looks like entirely on its own

Separately from any comparison to poetry, we also checked how the Qur'an's own internal structure holds together, with no poets involved at all, at three levels of granularity: the 114 individual chapters, the 60 traditional parts, and 30 pairs of consecutive parts.

Grouping	Clusters found	Left unclustered	Clustering quality (silhouette)
60 parts	3	0	0.882
30 part-pairs	3	0	0.543
114 chapters	9	3 chapters	0.663

All three cluster cleanly among themselves, with almost nothing left over. This says the Qur'an isn't a stylistically flat, uniform block of text; it has real, consistent internal structure, independent of anything to do with poetry at all.

Putting it together

No single result here settles the underlying question, and it shouldn't be read as if it does. What we have is a set of results that check each other.

The Qur'an, especially when treated as one combined text, does not read as maximally distinct from this poetry corpus by this measure. A specific, fairly small, recurring group of poets keeps showing up close to it, across several different ways of running the comparison.

Much of that closeness looks explainable by ordinary properties of the method rather than anything specific to the Qur'an. Minor poets with thin surviving data drift toward the center of the space. Heavily averaged combined vectors do the same thing, regardless of what goes into them. And a lone point of any kind, meaningless text included, tends to get pulled into whatever cluster is nearest rather than standing genuinely apart. These three effects, taken together, account for a large share of the numbers reported here.

At the same time, not everything reduces cleanly to those explanations. When the Qur'an is split into its 114 individual chapters and compared only against well-documented poets, the overlap disappears completely, a genuinely clean separation, and this is the strongest single result pointing toward the Qur'an and this poetry being stylistically distinct once the main confounds are controlled for. The fact that the whole-Qur'an-as-one-vector version of the same comparison didn't fully replicate that separation is itself informative: it suggests the earlier apparent closeness had more to do with how heavily averaged vectors behave in general than with anything particular about poetry.

We want to be direct about what this can and can't tell you. Every method used here measures how sentence embeddings sit relative to each other in a mathematical space. It does not measure, and cannot measure, who wrote anything. What's described above is a set of consistent findings about measurable writing style, nothing more and nothing less than that.